

Lesson 2 · Exponentials and Logarithms

NCEA Level 2 Mathematics · Algebra

By the end of today you can:

1. switch between power form and log form
2. use the three log laws to combine and solve
3. solve equations with the unknown in the power
4. answer growth and decay questions in context

Teaching line, not a question slide.

100 分钟。时间：Do Now 0:00–0:05；核心思想 0:05–0:07；Block 1 log 的意思 0:07–0:17；Block 2 log laws 0:17–0:32；Block 3 解指数方程 0:32–0:47；Block 4 增长和衰减 0:47–0:57；Classwork 0:57–1:17；真题演练 1:17–1:35；收尾 1:35–1:40。

发四份纸：Handout 2、Classwork 2、Practice set 2、Homework 2。学生需要计算器（log、ln 键）。

先用 5 分钟对 Homework 1 的答案（按 L 编号说错在哪），再开始 Do Now；时间不够就把 Block 4 的 Practice 放进作业。

Say it:

exponential = ek-spoh-NEN-shul

logarithm = LOG-uh-rith-um

ln = el-EN

end of notes

Do Now

- 1 Simplify $\frac{(2x)^3}{x^{-1}}$.
- 2 Write $\sqrt[3]{x^2}$ as a power of x .
- 3 Solve $2^x = 32$.
- 4 Factorise $2x^2 + 7x + 3$.
- 5 Which is bigger: 2^{10} or 10^3 ?

Q1 $8x^4$

Q2 $x^{\frac{2}{3}}$

Q3 $x = 5$

Q4 $(2x + 1)(x + 3)$

Q5 $2^{10} = 1024$ is bigger

Short solution:

Q1 $8x^3 \div x^{-1} = 8x^{3-(-1)}$

Q3 2, 4, 8, 16, 32: 数了 5 次

Watch for:

Q1 写 $8x^2$: 减负数 (L1)

Q4 $ac = 6$: 6 和 1

Diagnostic:

Q1、Q2 错: Lesson 1 的指数还不稳, Block 2 讲 log laws 前先回顾 index laws

Q3 靠猜也可以; 问他 “ $2^x = 30$ 怎么办”, 引出今天的 log

Q4 错: Block 3 的 disguised quadratic 要慢一点

end of notes

The one idea

$$2^3 = 8 \quad \Longleftrightarrow \quad \log_2 8 = 3$$

A log answers one question: **what power?**

$\log_2 8$ means: 2 to the power _____ gives 8.

❓ So what is $\log_2 16$?

Gap what (the power 3)

Q 4, because $2^4 = 16$

Short solution:

Q log 的答案就是一个 power; 把它读成问题: “2 的几次方是 16?”

Watch for:

答 8 ($16 \div 2$): 把 log 当成除法

Ask:

$\log_2 2$? (1) $\log_2 1$? (0)

end of notes

Quick review: powers

1 $2^6 =$ _____

2 $3^4 =$ _____

3 $10^{-2} =$ _____

4 $5^0 =$ _____

5

$$9^{\frac{1}{2}} = \underline{\hspace{2cm}}$$

Q1 64

Q2 81

Q3 0.01

Q4 1

Q5 3

Short solution:

Q3 $10^{-2} = \frac{1}{10^2} = \frac{1}{100}$

Q5 power $\frac{1}{2}$ 是 square root

Watch for:

Q3 答 -100 或 -20 : 负 index (L1)

讲义最后一页有常用的 power 表, 今天每个 log 都会用到

end of notes

Switching forms

$$y = b^x \iff \log_b y = x$$

Base to the bottom, y in the brackets.

$$3^4 = 81 \iff \log_3 ______ = ______$$

$$\log_5 x = 3 \iff x = ______$$

❓ In $\log_5 x = 3$, which number is the base?

Gap 81; $4; 5^3 = 125$

Q 5

Short solution:

Q base 是 log 下面的小数字；它在 power form 里是底数

Watch for:

写成 $x = 3^5$: base 和 power 弄反 (L11)

Say it:

base = BAYS

end of notes

When the unknown is the base

Solve $\log_x 49 = 2$.

$$x^2 = 49$$

$$x = 7 \quad \text{or} \quad x = -7 ?$$

❓ Can a base be negative? Which answer survives?

Q no; $x = 7$

Short solution:

Q base 必须是正数（且不是 1），所以 $x = -7$ 舍去，并写出理由

Watch for:

只写 $x = \pm 7$ 或只写 7 不说理由：评分表要 evidence of negative value being disregarded (L14)

end of notes

Logs you can do in your head

$$\log_7 1 = \underline{\hspace{1cm}} \quad \log_6 6 = \underline{\hspace{1cm}}$$

$$\log 1000 = \underline{\hspace{1cm}} \quad \ln e = \underline{\hspace{1cm}}$$

No base written: base 10 (the **log** key). $\ln$ is base $e \approx 2.718$ (the **ln** key).

❓ Why is $\log_7 1$ the same as $\log_6 1$?

Gap 0; 1; 3; 1

Q any base to the power 0 is 1

Short solution:

Q $7^0 = 1$, $6^0 = 1$, 所以都是 0

Watch for:

$\log 1000$ 答 100: 把 $\log$ 当成除以 10

计算器上 $\log$ 是 base 10, $\ln$ 是 base e ; 今天的指数方程两个都会用

end of notes

Practice

Q1 Evaluate (a) $\log_2 32$ (b) $\log_9 3$ (c) $\log 0.01$

Q2 Solve $\log_4 x = 3$.

Q3 Solve $\log_x 125 = 3$.

Q1 (a) 5; (b) $\frac{1}{2}$; (c) -2

Q2 $x = 64$

Q3 $x = 5$

Short solution:

Q1 (b) 9 的几次方是 3? square root, 所以 $\frac{1}{2}$ (u)

Q2 $x = 4^3$ (u)

Q3 $x^3 = 125$, cube root (u)

Watch for:

Q1 (b) 答 3: 方向反了 (L11)

Q3 这里 cube root 只有一个值, 不需要舍去

end of notes

The three log laws

$$\log_b(xy) = \log_b x + \log_b y \quad (\text{multiply: add the powers})$$

$$\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y \quad (\text{divide: subtract})$$

$$\log_b(x^n) = n \log_b x \quad (\text{power: multiply})$$

$$\log_2 3 + \log_2 5 = \log_2 \underline{\hspace{2cm}} \quad \log_2 12 - \log_2 4 = \log_2 \underline{\hspace{2cm}}$$

❓ Which index law is behind the first log law?

Gap 15; 3

Q $x^a \times x^b = x^{a+b}$: multiplying adds the powers

Short solution:

Q log 就是 power, 所以 log 的加法对应里面的乘法
三条都在 formula sheet 上, 不用背, 但要会用

Watch for:

$\log_2 3 + \log_2 5 = \log_2 8$: 加法对应乘法, 不是加法 (L12)

end of notes

Writing a single log

Write $2 \log x + \log 5 - \log 10$ as a single log.

$$= \log x \text{---} + \log 5 - \log 10$$

$$= \log \frac{5x^2}{10} = \log \text{---}$$

❓ Why move the 2 up as a power first?

Gap 2; $\frac{x^2}{2}$

Q the add and subtract laws only work on logs with nothing in front

Short solution:

Q 前面的系数先变成 power, 然后加号进分子、减号进分母, 最后约分

Watch for:

写成 $\log 2x$: 2 没有变成 power (L13)

答案停在 $\log \frac{5x^2}{10}$: 分数要约到最简

end of notes

Equations with two logs

Solve $\log_3(x+4) + \log_3(x-4) = 2$.

$$\log_3((x+4)(x-4)) = 2$$

$$x^2 - 16 = 3^2$$

$$x = 5 \quad \text{or} \quad x = -5$$

❓ Put $x = -5$ into $\log_3(x-4)$. What goes wrong?

Q $\log_3(-9)$ does not exist: reject $x = -5$; $x = 5$

Short solution:

Q 先合并成一个 log, 再换成 power form: $x^2 - 16 = 9$

Watch for:

两个解都留下 (L14)

右边的 2 换成 $3^2 = 9$, 不是 2^3 (L11)

end of notes

Spot the error (1)

Sam's answer has 2 errors. Find each one, fix it, and say why it loses marks.

Sam's answer

(a) Expand $\log(x + 3)$ $\log(x + 3) = \log x + \log 3$

(b) Write $3 \log 2$ as a single log $3 \log 2 = \log 6$

1. (a) there is no law for $\log(x + 3)$: it cannot be split (L12)

2. (b) $3 \log 2 = \log 2^3 = \log 8$, not $\log 6$ (L13)

Short solution:

(a) $\log(x + 3)$ already simplest

(b) $\log 8$

Watch for:

(a) 代 $x = 7$ 检查: $\log 10 = 1$, 但 $\log 7 + \log 3 = \log 21 \approx 1.32$

end of notes

Practice

Q1 Write $\log(4a) + 3\log\left(\frac{a}{2}\right)$ as a single logarithm.

Q2 Solve $\log_2(x+2) + \log_2(x-2) = 5$.

Q3 Solve $\log x - \log(x-3) = \log 4$.

Q1 $\log \frac{a^4}{2}$

Q2 $x = 6$

Q3 $x = 4$

Short solution:

Q1 $\log 4a + \log \frac{a^3}{8} = \log \frac{4a^4}{8}$ (r)

Q2 $x^2 - 4 = 32$, $x = \pm 6$, 舍去 -6 (r)

Q3 $\log \frac{x}{x-3} = \log 4$, 所以 $x = 4x - 12$ (u)

Watch for:

Q1 $\left(\frac{a}{2}\right)^3$ 分母也要立方 (L2)

Q2 不舍去 -6 (L14)

end of notes

The unknown in the power

Solve $3^x = 50$.

$$\log 3^x = \log 50$$

$$x \log 3 = \log 50$$

$$x = \frac{\log 50}{\log 3} = \underline{\hspace{2cm}}$$

❓ Why can't you just divide 50 by 3?

Gap 3.561

Q 3^x is not $3 \times x$; the power has to come down first

Short solution:

Q 两边取 log, power law 把 x 拿下来, 再除以 $\log 3$

检查: $3^3 = 27$, $3^4 = 81$, 所以答案在 3 和 4 之间

Watch for:

$$\frac{\log 50}{\log 3} \text{ 写成 } \log \frac{50}{3} \text{ (L12)}$$

end of notes

Bracket the power

Solve $5^{2x-1} = 200$.

$$(2x - 1) \log 5 = \log 200$$

$$2x - 1 = \frac{\log 200}{\log 5} = 3.2920 \dots$$

$$x = \underline{\hspace{2cm}}$$

❓ What goes wrong if you write $2x - 1 \log 5$?

Gap 2.146

Q only the 1 gets multiplied by $\log 5$

Short solution:

Q 整个 power $(2x - 1)$ 拿下来要加括号；先求 $2x - 1$ ，最后再解 x
中间值保留计算器里的全部位数，最后才四舍五入（L16）

Watch for:

没有括号（L13）

end of notes

Same base: solve exactly

Solve $4^{x+1} = 8^x$.

$$4 = 2^2 \quad \text{and} \quad 8 = 2^3$$

$$2^{2(x+1)} = 2^{3x}$$

$$2x + 2 = 3x$$

② What base do 4 and 8 share? What is x ?

Gap 3

Q base 2; $x = 2$

Short solution:

Q 底数相同时 power 相等；精确解，不用计算器

检查： $4^3 = 64 = 8^2$

Watch for:

$(2^2)^{x+1}$ 写成 2^{2x+1} ：括号（L1 的 power of a power）

end of notes

e and $\ln$

Solve $e^{0.3t} = 5$.

$$\ln e^{0.3t} = \ln 5$$

$$0.3t = \ln 5$$

$$t = \underline{\hspace{2cm}}$$

❓ What undoes e to a power?

Gap 5.365

Q $\ln$, because $\ln e^k = k$

Short solution:

Q $t = \frac{\ln 5}{0.3}$; 也可以用 $\log$: $t = \frac{\log 5}{0.3 \log e}$, 答案一样

Watch for:

写 $t = \ln \frac{5}{0.3}$: 先除再取 $\ln$ 是错的

考试里常见 $T = T_0 + ke^{-rt}$ 的冷却模型, 先把 e 的那一项单独移出来

end of notes

A quadratic in disguise

Solve $3^{2x} - 4 \times 3^x + 3 = 0$.

Let $u = 3^x$. Then $3^{2x} = (3^x)^2 = u^2$.

$$u^2 - 4u + 3 = 0$$

$$(u - 1)(u - 3) = 0$$

$$3^x = 1 \quad \text{or} \quad 3^x = 3$$

② What are the two values of x ?

Q $x = 0$ or $x = 1$

Short solution:

Q $3^x = 1$ 时 $x = 0$; $3^x = 3$ 时 $x = 1$ (t)

如果某个 u 是负数, 要舍去: 3^x 永远是正的

Watch for:

解出 u 就停: 要换回 x

认出 $3^{2x} = (3^x)^2$ 是 Excellence 的关键一步

end of notes

Spot the error (2)

Solve $5^{x-2} = 40$.

Sam's answer has 2 errors. Find each one, fix it, and say why it loses marks.

Sam's answer

$$x - 2 \log 5 = \log 40$$

$$x - 2 = \frac{\log 40}{\log 5} = \log 8 = 0.903$$

$$x = 2.903$$

1. the power comes down in a bracket: $(x - 2) \log 5 = \log 40$ (L13)

2. $\frac{\log 40}{\log 5} \neq \log 8$: divide on the calculator, $= 2.292$ (L12)

Short solution:

$$x - 2 = 2.292, \quad x = 4.292$$

Watch for:

检查: $5^{4.292-2} = 5^{2.292} \approx 40$; Sam 的 $5^{0.903} \approx 4.3$

end of notes

Practice

Q1 Solve $2^x = 300$.

Q2 Solve $7^{3x+1} = 90$.

Q3 Solve $9^x = 27^{x-1}$ exactly.

Q1 $x = 8.229$

Q2 $x = 0.4375$

Q3 $x = 3$

Short solution:

Q1 $x = \frac{\log 300}{\log 2}$ (u)

Q2 $3x + 1 = \frac{\log 90}{\log 7} = 2.3124$ (u)

Q3 $3^{2x} = 3^{3x-3}$, $2x = 3x - 3$ (r)

Watch for:

Q2 括号 (L13)

Q3 用计算器得到小数也对，但题目要 exactly

end of notes

Growth and decay

$$P = Ar^t$$

A car bought for \$18 000 loses 15% of its value each year.

$$V = 18\,000 \times \text{_____}^t$$

❓ Why is the growth factor 0.85 and not 0.15?

Gap 0.85

Q each year it keeps 85% of its value

Short solution:

Q 增长 $R\%$: 乘 $1 + \frac{R}{100}$; 减少 $R\%$: 乘 $1 - \frac{R}{100}$

Watch for:

用 0.15: 一年后只剩 15% (L15)

end of notes

When does it happen?

When is the car first worth less than \$6000?

$$18\,000(0.85)^t = 6000$$

$$0.85^t = \frac{1}{3}$$

$$t = \frac{\log \frac{1}{3}}{\log 0.85} = 6.76$$

❓ Is the answer “6.76”? What do you write?

Q no: it is first worth less than \$6000 during the 7th year, after about 6.8 years

Short solution:

Q 先除以 18000，再取 log；最后一句话要有单位和情境

Watch for:

只写 6.76 或 “6 years”：没解释 (L16)

$\log \frac{1}{3}$ 和 $\log 0.85$ 都是负数，相除是正数

end of notes

What rate?

\$5000 grows to \$7500 in 8 years at a fixed annual rate $R\%$.

$$5000 \left(1 + \frac{R}{100}\right)^8 = 7500$$

$$1 + \frac{R}{100} = 1.5^{\frac{1}{8}} = \underline{\hspace{2cm}}$$

② When the unknown is the base, do you take logs or a root?

Gap 1.0520

Q a root: the 8th root, then $R = 5.2\%$

Short solution:

Q 未知数在 base, 不在 power: 开 8 次方 (power $\frac{1}{8}$)

Watch for:

答 $R = 1.052$ 或 0.052 : 要换成百分数

答 $R = 6.25$ (把 50% 平均分给 8 年): 复利不是平均

end of notes

Spot the error (3)

A \$15 000 car loses 20% of its value each year. When is it first worth less than \$5000?

Sam's answer has 2 errors. Find each one, fix it, and say why it loses marks.

Sam's answer

$$15000(0.2)^t = 5000 \quad 0.2^t = \frac{1}{3} \quad t = \frac{\log(1/3)}{\log 0.2} = 0.683$$

Answer: 0.683

1. a 20% loss keeps 80%: the factor is 0.8, not 0.2 (L15)
2. the answer needs units and a sentence in context (L16)

Short solution:

$$0.8^t = \frac{1}{3}, \quad t = 4.92: \text{ during the 5th year, after about 4.9 years}$$

Watch for:

Sam 的 0.683 年明显不合理：一年后车还值 \$12000

end of notes

Q1 A bacteria count is $N = 200(1.35)^t$ after t hours. When does it first pass 10 000?

Q2 A coffee cools as $T = 20 + 65e^{-0.08t}$ (t in minutes). When is it 50°C ?

Q3 A town grows from 4200 to 5600 people in 6 years at a constant rate. Find the annual rate.

Q1 after about 13.0 hours

Q2 after about 9.7 minutes

Q3 about 4.9% per year

Short solution:

$$\text{Q1 } 1.35^t = 50, \quad t = \frac{\log 50}{\log 1.35} = 13.04 \text{ (u)}$$

$$\text{Q2 } 65e^{-0.08t} = 30, \quad -0.08t = \ln \frac{30}{65}, \quad t = 9.665 \text{ (r)}$$

$$\text{Q3 } r^6 = \frac{4}{3}, \quad r = 1.0491 \text{ (r)}$$

Watch for:

Q2 先减 20 再除 65: 顺序反了得不到 e 单独一项

每题最后写一句话 (L16)

end of notes

Classwork 2 · about 20 minutes

- **Part A:** switching forms, single logs, solving
- **Part B:** two-log equations and a decay model
- **Part C:** a quadratic in disguise, explained
- **Extension**, if you finish

Teaching line, not a question slide.

答案在 classwork2_answers.pdf (同版面, 红色答案)。

A1 5; A2 36; A3 $\log 10 = 1$; A4 $x = 2.410$; A5 $x = 2$

B1 $x = 2.548$; B2 $x = 4$ (舍去 -2) ; B3 during the 7th year

C: $x = 0$ or $x = 2$

Extension: $x = 2.710$

学生做的时候不要讲; 收卷后先看 Part C 有没有写 $u = 2^x$ 和舍去的理由

end of notes

(c) Jessica is investigating a compounding investment. She wants to know how long it would take for an investment of \$1000 to double in value to \$2000. She forms the following equation:

$$2000 = 1000 \left(1 + \frac{R}{100} \right)^D$$

where R is the rate of return on the investment, as a percentage, and D is the time that the investment would take to double in value, in years.

(i) If an investment takes 11 years to double in value, what is its rate of return?

(i) $R = 6.5\%$

Short solution:

(i) $\left(1 + \frac{R}{100} \right)^{11} = 2$, $1 + \frac{R}{100} = 2^{\frac{1}{11}} = 1.065$

评分：列出方程并用 11 次方根是 u，正确答案是 r

Watch for:

未知数在 base：开方，不是取 log

end of notes

(ii) By making D the subject of the expression:

$$2000 = 1000 \left(1 + \frac{R}{100} \right)^D$$

show that

$$D = \frac{\log(2)}{\log\left(1 + \frac{R}{100}\right)}$$

(ii) divide by 1000, take logs of both sides, bring D down, divide

Short solution:

$$(ii) \left(1 + \frac{R}{100} \right)^D = 2$$

$$\log \left(1 + \frac{R}{100} \right)^D = \log 2$$

$$D \log \left(1 + \frac{R}{100} \right) = \log 2$$

评分：写出两边取 log 的一行（第 2 或第 3 行）是 u；完整推出给定式子是 r

Watch for:

Show that 题只写最后一行：没有分

end of notes

In her research, Jessica comes across a simple but **approximate** rule for calculating D , the time that the investment would take to double in value. It is commonly called the ‘Rule of 72’, and it states

that: $D = \frac{72}{R}$

Jessica wonders how close the values of D from the ‘Rule of 72’ are to those calculated using the actual expression, which is:

$$D = \frac{\log(2)}{\log\left(1 + \frac{R}{100}\right)}$$

Teaching line, not a question slide.

这是 (iii) 的背景，先读题，下一页是问题

问学生： $R = 8$ 时 Rule of 72 给几年？（9 年）真实值？ $\left(\frac{\log 2}{\log 1.08} = 9.006 \text{ 年}\right)$

Say it:

approximate = uh-PROK-sih-mit

end of notes

(iii) Show clearly that the value of R for which the ‘Rule of 72’ exactly calculates D , is the solution to the equation:

$$2^R - \left(1 + \frac{R}{100}\right)^{72} = 0$$

You do not need to solve this equation.

(iii) set the two expressions for D equal and rearrange with log laws

Short solution:

$$(iii) \frac{72}{R} = \frac{\log 2}{\log\left(1 + \frac{R}{100}\right)}$$

$$72 \log\left(1 + \frac{R}{100}\right) = R \log 2$$

$$\log\left(1 + \frac{R}{100}\right)^{72} = \log 2^R, \text{ 所以 } \left(1 + \frac{R}{100}\right)^{72} = 2^R$$

评分：两式相等是 u；处理 power（第 3 行）是 r；推出给定式子且数学表述正确是 t

Watch for:

两边的 log 去掉之前要先写成 $\log(\dots) = \log(\dots)$ 的形式

end of notes

(a) (ii) Solve the equation $2^x = 2022$.

(ii) $x = 10.98$

Short solution:

(ii) $x \log 2 = \log 2022$; 评分: Correct solution (u); 写 $\log_2 2022$ 也接受

Watch for:

检查: $2^{11} = 2048$, 所以略小于 11

end of notes

(b) Simplify the following expression fully, writing your answer as a single logarithm.

$$\log(3a) + 2 \log\left(\frac{a}{6}\right)$$

(b) $\log \frac{a^3}{12}$

Short solution:

(b) $\log 3a + \log \frac{a^2}{36} = \log \frac{3a^3}{36}$

评分：分数没有化简是 u；正确的 single log 是 r

Watch for:

$\left(\frac{a}{6}\right)^2$ 分母是 36 (L2)

end of notes

(c) Consider the equation $\log_2(x - a) - \log_2(x + a) = c$, where a and c are constants.

(i) Show that when x is made the subject of this equation,

$$x = a \frac{1 + 2^c}{1 - 2^c}$$

Ensure that you use correct mathematical statements in your reasoning.

(i) combine the logs, switch to power form, collect the x terms, factorise

Short solution:

(i) $\log_2 \frac{x-a}{x+a} = c$, 所以 $\frac{x-a}{x+a} = 2^c$

$$x - a = 2^c x + 2^c a, \quad x(1 - 2^c) = a(1 + 2^c)$$

评分：合并 log 是 u；写出 power form（第 3 行）是 r；完整正确的推理是 t

Watch for:

这是 Lesson 1 的“换主元”： x 在两边，要提公因式（L10）

end of notes

(ii) The equation $\log_2(x - a) - \log_2(x + a) = c$ is only possible to solve for some values of a and for some values of c .

Explaining your reasoning clearly, describe which values of a and of c will make the equation possible to solve.

You may find it useful to recall that, when x is made the subject of this equation,

$$x = a \frac{1 + 2^c}{1 - 2^c}$$

(ii) $c \neq 0$ and $a \neq 0$, with $x - a > 0$ and $x + a > 0$; this happens exactly when a and c have opposite signs

Short solution:

(ii) $c = 0$ 时 $1 - 2^c = 0$, x 没有定义

$a = 0$ 时 $x = 0$, $\log_2 0$ 没有定义

两个 $\log$ 的括号都要为正

评分：一个条件加理由是 r ；两个条件加理由是 t

Watch for:

只写条件不写理由：Explaining your reasoning clearly

进一步： $a > 0$ 需要 $c < 0$ ； $a < 0$ 需要 $c > 0$ （评分表只要求前面的条件）

end of notes

Ways to lose marks

- **L11** Base to the bottom, y in the brackets
- **L12** No law for $\log(x + 3)$; $\frac{\log 40}{\log 5}$ is not $\log 8$
- **L13** $(x - 2) \log 5$: the whole power in a bracket
- **L14** Check every answer: logs need positive brackets
- **L15** A 20% loss multiplies by 0.8
- **L16** A sentence with units; round at the end

Teaching line, not a question slide.

讲义上有 L11–L16 的完整说明

让学生对照 Classwork 和 Practice set 的错，说出是哪一个 L

end of notes

Summary

- A log is a power: $\log_b y = x$ means $b^x = y$.
- Laws: plus multiplies, minus divides, a number in front becomes a power.
- Unknown in the power: take logs, bracket the power.
- Growth: $P = Ar^t$; answer in a sentence with units.

Teaching line, not a question slide.

让学生用自己的话说第一条：“log 问的是什么？”

end of notes

Homework 2 · about 60 minutes

- **Quick review:** four short items
- **Question One:** meaning of logs and the log laws
- **Question Two:** exponential equations
- **Question Three:** growth and decay in context

Each question is marked like the real paper: N0 to E8.

Teaching line, not a question slide.

答案在 hw2_answers.pdf; 每题的 (e) 是 Excellence 题
下节课开头 5 分钟对答案

end of notes